



# MindHx

*Notice the signal. Keep the human.*

An early-detection mental-health triage aid that fuses voice, language, and validated clinical questionnaires into one explainable, weighted risk score — private by default, bilingual (English/Urdu) throughout, never a diagnosis.

Next.js 16

FastAPI

Qwen / DashScope

PHQ-9 · GAD-7 · K10

English + اردو

# Lower the barrier *to a first screening step.*

Depression, anxiety, and related conditions are frequently under-screened, especially where access to mental-health professionals is limited and stigma discourages self-report. MindHx's objective is a private, low-friction check-in that surfaces a risk signal a person can act on — without pretending to be a diagnosis, a replacement for a clinician, or a clinically calibrated instrument.

## Multimodal, not single-signal

Voice, text, and three validated questionnaires are fused into one score, with a genuine per-signal attribution breakdown shown on the results page — not an opaque number.

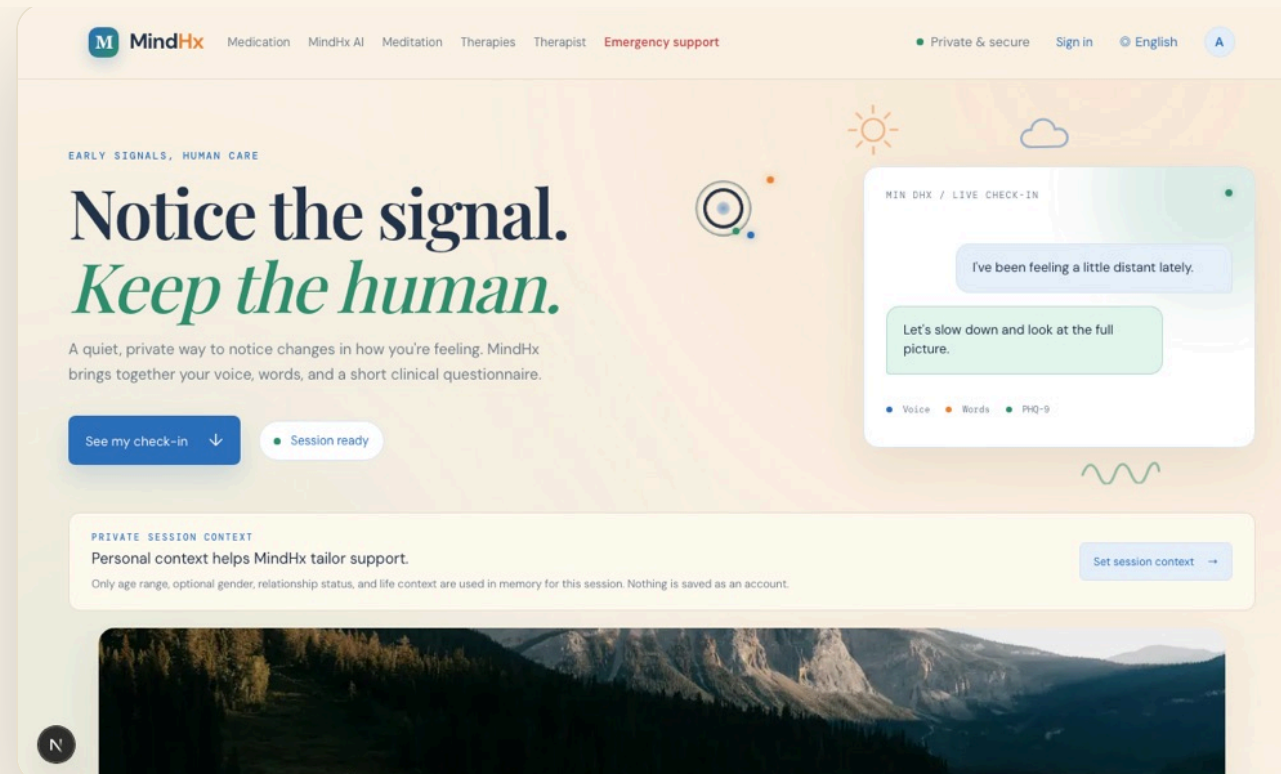
## Route, don't diagnose

Every page is worded in risk-tier language, never diagnostic language. PHQ-9 item 9 and self-harm language short-circuit everything else and go straight to Emergency Support.

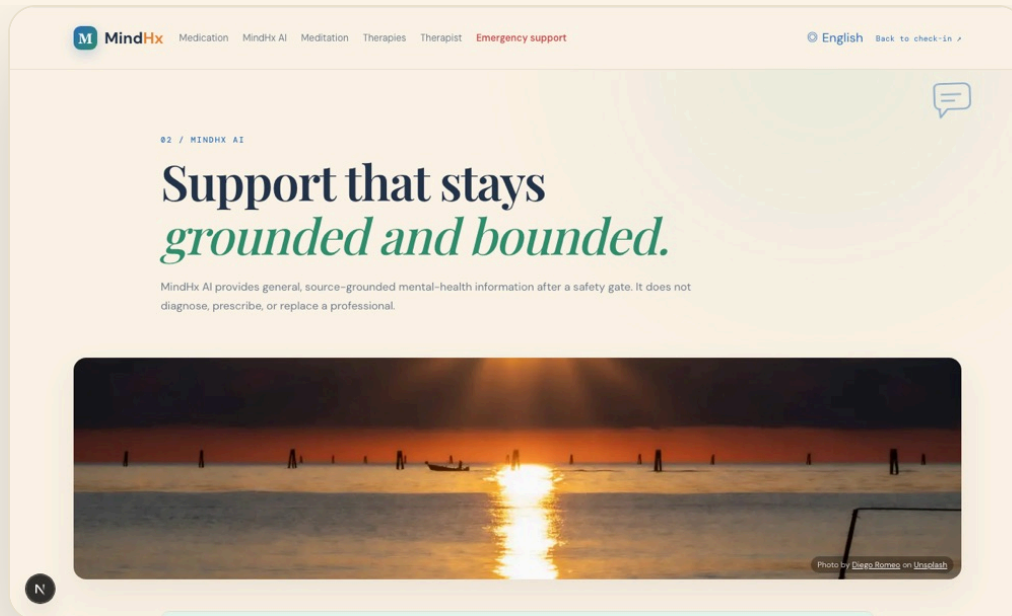
## Minimal data, accounts optional

The core check-in never requires signing in. Audio is processed and discarded. An optional account stores only score, band, and themes — never a transcript.

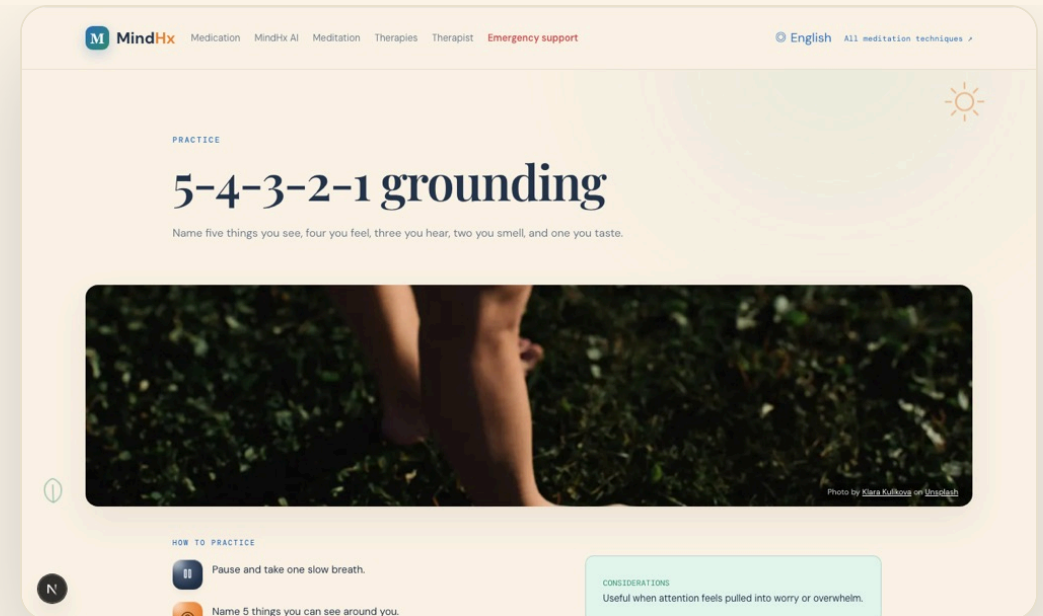
# Voice, words, and a *short clinical questionnaire* — in one flow.



# A safety-gated AI chat, *and a practice library with* step-by-step 3D icons.

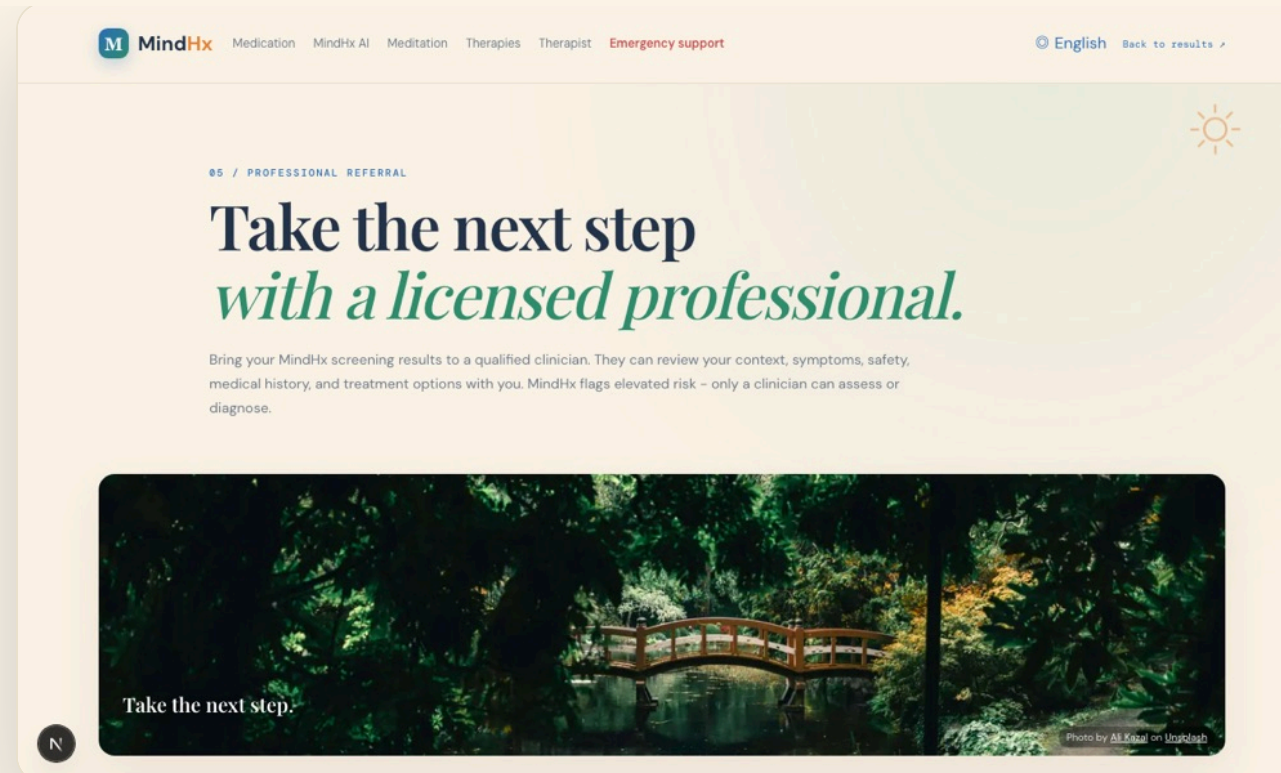


MindHx AI — bounded, source-grounded chat



5-4-3-2-1 grounding — original 3D-shaded step icons

# A real Pakistan *therapist directory* — not a static, stale list.



# Every page, in English *and full RTL Urdu* — not a separate build.

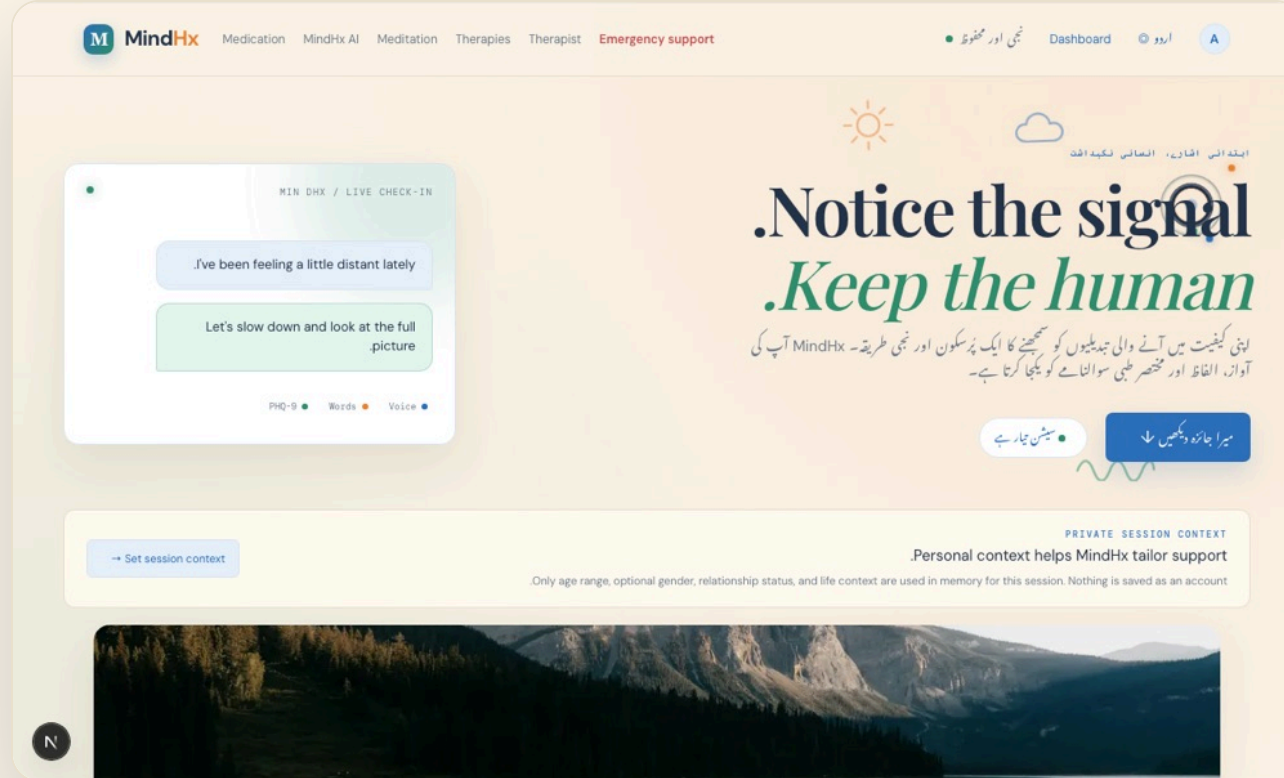
Each page ships one bilingual copy object; a single header toggle switches language and text direction together. The screenshots on the following pages are the same live app, the same session, with only the language toggle clicked.

Header-level toggle on every page

Full RTL layout mirroring

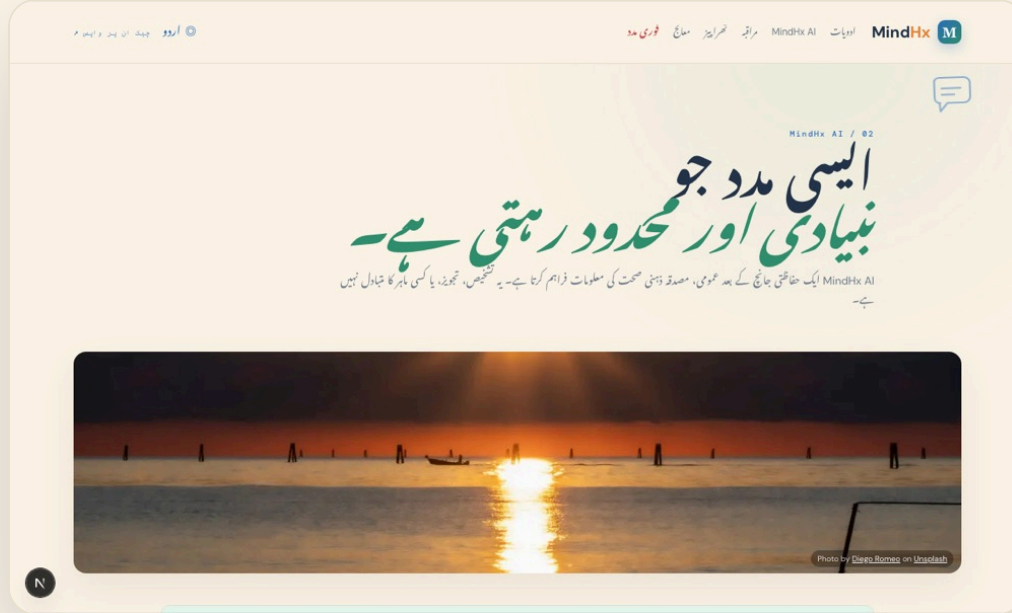
Server-side Urdu-script guard on AI replies

# آہستہ چلیں۔ اشارے کو محسوس کریں۔

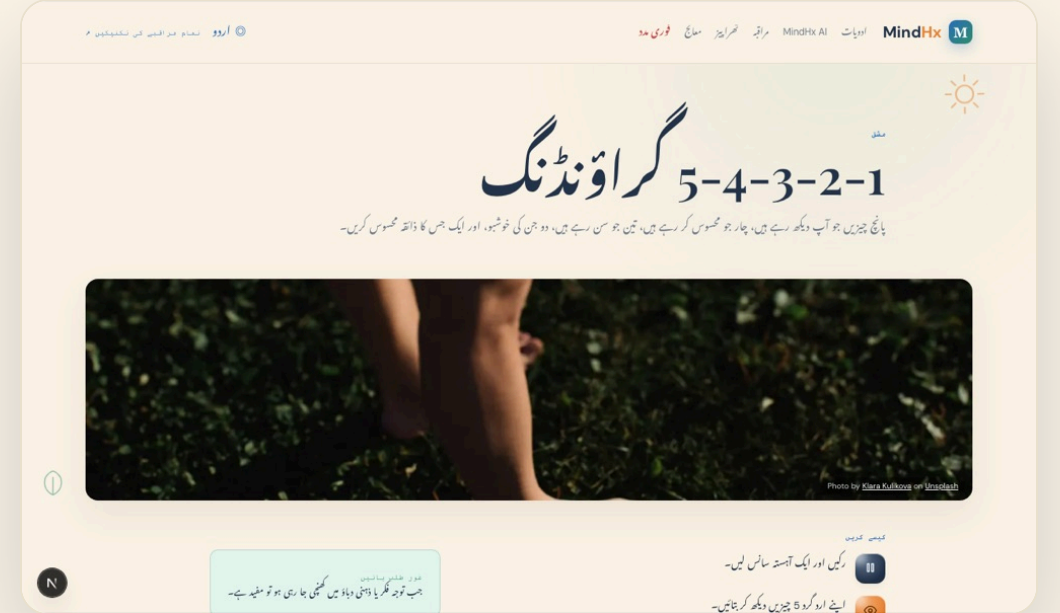




# MindHx AI اور مراقبہ کی تکنیکیں



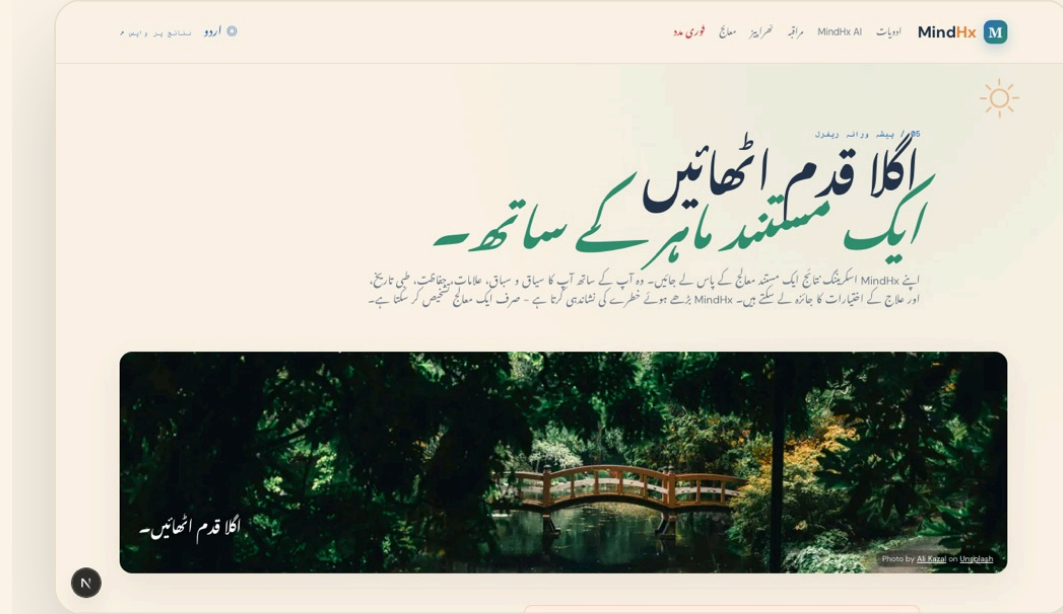
اردو - MindHx AI



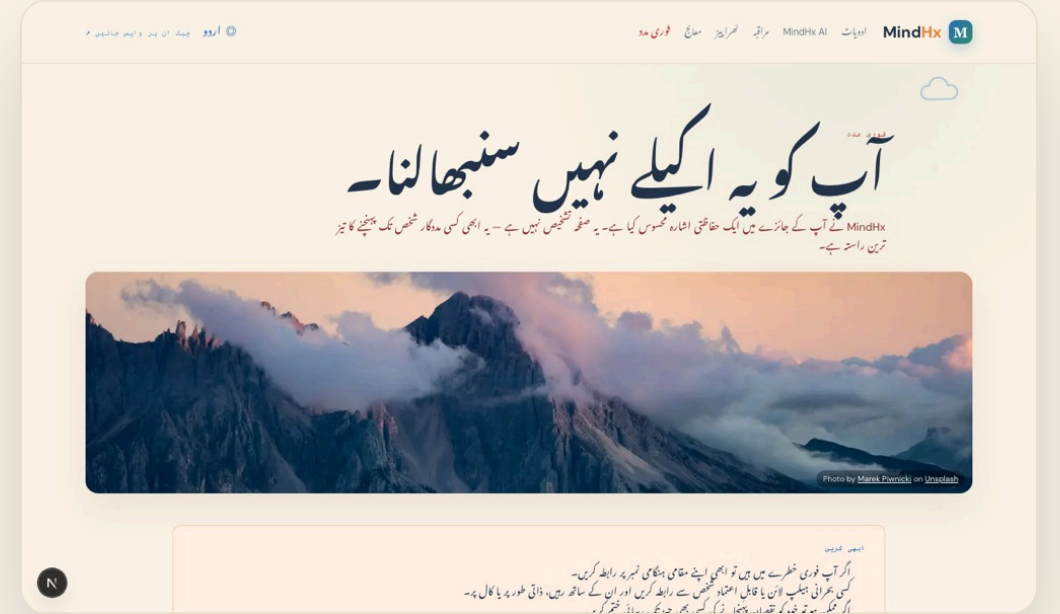
گراؤنڈنگ - اردو 5-4-3-2-1



# معالج تلاش کریں اور فوری مدد



معالج تلاش کریں - اردو



فوری مدد - اردو

# What's built — *and what's deliberately not.*

## BUILT

- Voice + text + PHQ-9/GAD-7/K10 fused into one explainable score
- Bounded, source-grounded AI chat with a hard safety gate
- Original 3D step-icon library across every technique/therapy page
- Full English/Urdu bilingual UI with RTL layout
- Real Pakistan therapist/psychiatrist directory
- Optional accounts — bcrypt + JWT, aggregate-only history

## NOT YET BUILT — DELIBERATELY

- **Score calibration** — no labeled outcome data yet; faking it would make the claim false
- **Clinically validated Urdu PHQ-9/GAD-7/K10** — current translation is demo-only
- **Real acoustic biomarker vendor** — today's signal is an explicit heuristic proxy
- **Auth hardening** — rate-limiting and httpOnly cookies still pending

# Two services.

## *Stateless by default.*

A Next.js 16 (App Router) frontend and a FastAPI backend. The screening/risk-assessment endpoints are fully stateless — every request self-contained, nothing persisted. An optional layer adds PostgreSQL-backed accounts purely for people who choose to save history.

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BACKEND ENDPOINTS

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FRONTEND ROUTES

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BACKEND TESTS

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FUSED SIGNALS

# Selected backend endpoints

| Endpoint                                  | Purpose                                                                                                                        |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| POST /analyze-voice                       | Heuristic prosodic signal (pause ratio, loudness variability, speaking rate) — pluggable via VOICE_BIOMARKER_PROVIDER          |
| POST /analyze-text                        | Sentiment + crisis-language detection — Qwen (DashScope) → OpenRouter → local heuristic fallback                               |
| POST /ai/chat                             | Bounded, source-grounded chat; safety-gated before any content is returned; server-side Urdu-script guard on generated replies |
| POST /risk-assess                         | Fuses all signals into the combined score, band, explanation, and routing decision                                             |
| POST /score-phq9, /score-gad7, /score-k10 | Individual validated-questionnaire scoring                                                                                     |
| POST /auth/register, /auth/login          | Optional account creation/sign-in — bcrypt + JWT                                                                               |
| POST /checkins, GET /checkins             | Save/list a signed-in user's check-in history (aggregate results only, never a transcript)                                     |
| POST /synthesize                          | Urdu text-to-speech via Uplift AI                                                                                              |

# From a private check-in *to a real next step.*

## In-app crisis routing

PHQ-9 item 9 or detected self-harm language routes straight to Emergency Support — before any score is computed.

## Bounded AI chat

Answers general questions from a fixed, reviewed reference library; escalates rather than engages on any safety concern.

## Resource library

Medication reference, meditation techniques, and therapy approaches — framed as general education, never personalized treatment.

## Optional dashboard

For signed-in users: a history of past scores/bands/themes to track change over time — never a diagnosis, never raw content.

# Compressing a screening step *into a few private minutes.*

- A full check-in — voice, free-text, three questionnaires — takes **under 5 minutes** and requires no account.
- The results page gives a person something concrete for a first clinician conversation, rather than starting from zero.
- Once labeled outcome data exists, **calibration work** (Platt scaling / isotonic regression) would let the score become an actual probability — a prerequisite for any clinical claim.
- A **validated Urdu instrument** is the single highest-leverage localization gap — it would make MindHx usable as a real screening tool in Urdu-speaking clinical settings, not just a bilingual demo.
- A real acoustic biomarker vendor would upgrade the voice signal from a heuristic proxy to a validated one.



Notice the signal.  
*Keep the human.*

MindHx is a screening and triage aid, not a diagnosis. Every score is a starting point for a conversation with a qualified professional — never a replacement for one.